StreamingApp – Container Orchestration Assignment

GitHub Repository

Repository Link: https://github.com/dhikalerahulrd-a11y/StreamingApp.git

Project

StreamingApp – Container Orchestration using Docker, Kubernetes, Helm and AWS EKS.

Technologies

- Docker
- Kubernetes
- Helm
- Jenkins
- Amazon EKS
- Amazon ECR
- MongoDB
- Ingress
- CloudWatch

Completed Components

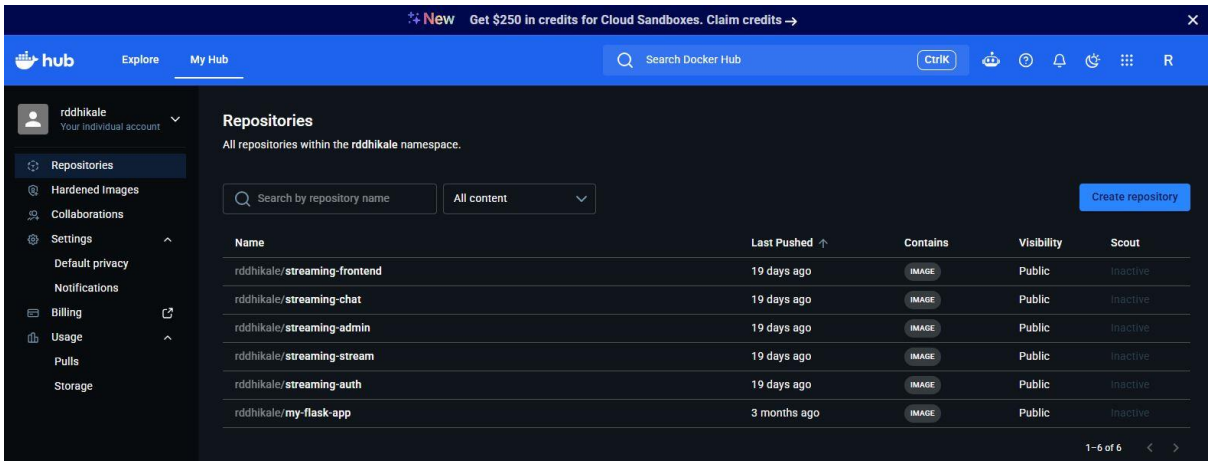
- Containerized 5 application services
- Docker images created and pushed
- Kubernetes deployments and services configured
- MongoDB StatefulSet and persistent storage configured
- Helm chart created
- Ingress configured
- Application deployed to AWS EKS
- Scaling demonstrated
- Rolling update demonstrated
- Pod self-healing verified
- Application smoke tests completed
- Jenkins CI pipeline configured
- CloudWatch monitoring/logging configured
- Project documentation uploaded to GitHub

Screenshots:

1. Docker

```
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp> Docker images
b17flasksamplems2:v1                                                  6c533e6a2ef6       110MB         26.9MB
demoapp:v1.0                                                          01c68885708a       110MB         26.9MB    U
democs-app:v1                                                         985a15441523       219MB         53.7MB
docker/desktop-cloud-provider-kind:v0.6.0                             46d51b990e97       613MB          168MB    U
docker/desktop-cloud-provider-kind:v0.7.0                             f078b9ccf478       628MB          171MB
docker/desktop-containerd-registry-mirror:v0.0.4                      1de142695bb0       52.9MB        16.9MB    U
ecs-basics-app:latest                                                 8cc8071a42fc       219MB         53.7MB
envoyproxy/envoy:v1.36.7                                              29d778ba078e       253MB         63.9MB    U
flask-app:latest                                                      52eac0b81536       199MB         48.3MB    U
flask-app:v1                                                          ed6bf313b25e       110MB         26.9MB    U
flask-app:v2                                                          f7b02ca3bb9a       110MB         26.9MB    U
flask-practice:test                                                   324b031bf73e       281MB         66.2MB    U
httpd:latest@sha256:979c38c2228d28c2edfd45c6e27dcee1c7b4a101a5526721ae8ece454e89e99e   979c38c2228d       45.3MB        45.3MB
kindest/node:v1.36.1                                                  3489c7674813       1.31GB         383MB    U
moby/buildkit:buildx-stable-1                                         2f5adac4ecd1       356MB          111MB    U
mongo:6                                                               8b6d8f5bbedb       1.1GB          282MB    U
mongo:latest                                                          951c2ff9fc6b       1.3GB          339MB    U
nginx:latest                                                          5a88c9c45479       241MB           66MB    U
public.ecr.aws/y9d2a0u0/ecs-basics-app:latest                         8cc8071a42fc       219MB         53.7MB
public.ecr.aws/y9d2a0u0/rahul/demoapp:dev                             ed6bf313b25e       110MB         26.9MB    U
public.ecr.aws/y9d2a0u0/rahul/demoapp:v2                              ed6bf313b25e       110MB         26.9MB    U
python:3.11.15-alpine3.23                                             f73754c398b2       84.3MB          21MB
rahul/demoapp:latest                                                  12c115edb1e0       110MB         26.9MB    U
rddhikale/streaming-admin:1.0.0                                       f170f38acc98       272MB         59.6MB
rddhikale/streaming-auth:1.0.0                                        e4010d90632a       241MB         56.4MB
rddhikale/streaming-chat:1.0.0                                        fdb363c6cc2a       230MB         54.8MB
rddhikale/streaming-frontend:1.0.0                                    290008356ded       78.2MB        21.9MB
rddhikale/streaming-stream:1.0.0                                      d59c84e32b40       303MB         74.8MB
rddhikale90/ecs-demo-app:v1                                           985a15441523       219MB         53.7MB
rddhikale90/my-flask-app:v1                                           ed6bf313b25e       110MB         26.9MB    U
rddhikale90/my-flask-app:v2                                           f7b02ca3bb9a       110MB         26.9MB    U
streamingapp-admin:latest                                             4cf921799124       325MB         64.9MB    U
streamingapp-auth:latest                                              3d20f130ac67       241MB         56.4MB    U
streamingapp-chat:latest                                              0629d9597cd3       261MB         59.1MB    U
streamingapp-frontend:latest                                          60fb609da16e       78.2MB        21.9MB    U
streamingapp-streaming:latest                                         50fb8e8c0b09       355MB         79.8MB    U
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp>
```

2. Docker Hub



3. EKS

```
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp> kubectl get nodes
NAME                                           STATUS   ROLES    AGE     VERSION
ip-192-168-0-132.ap-south-1.compute.internal   Ready    <none>   3h36m   v1.34.11-eks-3b4a6ca
ip-192-168-12-93.ap-south-1.compute.internal   Ready    <none>   68m     v1.34.11-eks-3b4a6ca
ip-192-168-44-58.ap-south-1.compute.internal   Ready    <none>   54m     v1.34.11-eks-3b4a6ca
ip-192-168-57-147.ap-south-1.compute.internal  Ready    <none>   3h36m   v1.34.11-eks-3b4a6ca
ip-192-168-62-237.ap-south-1.compute.internal  Ready    <none>   3h36m   v1.34.11-eks-3b4a6ca
ip-192-168-64-76.ap-south-1.compute.internal   Ready    <none>   68m     v1.34.11-eks-3b4a6ca
ip-192-168-72-165.ap-south-1.compute.internal  Ready    <none>   3h36m   v1.34.11-eks-3b4a6ca
ip-192-168-85-204.ap-south-1.compute.internal  Ready    <none>   29m     v1.34.11-eks-3b4a6ca
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp>
```

4. Pods

```
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp> kubectl get pods -n streaming
NAME                                  READY   STATUS    RESTARTS   AGE
mongo-0                               1/1     Running   0          39m
streamingapp-admin-74d84b56c6-vjtlg   1/1     Running   0          16m
streamingapp-auth-6bb6f6c4c5-qrtmv    1/1     Running   0          16m
streamingapp-chat-6d6b6bb854-kvpx6    1/1     Running   0          16m
streamingapp-frontend-67794466c6-kwxsq  1/1   Running   0          79m
streamingapp-streaming-564f6dbd95-hglmw  1/1  Running   0          16m
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp>
```

5. Services

```
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp> kubectl get svc -n streaming
NAME        TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)     AGE
admin       ClusterIP   10.100.197.131   <none>        3003/TCP    80m
auth        ClusterIP   10.100.181.205   <none>        3001/TCP    80m
chat        ClusterIP   10.100.75.16     <none>        3004/TCP    80m
frontend    ClusterIP   10.100.182.142   <none>        80/TCP      80m
mongo       ClusterIP   None             <none>        27017/TCP   80m
streaming   ClusterIP   10.100.93.128    <none>        3002/TCP    80m
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp>
```

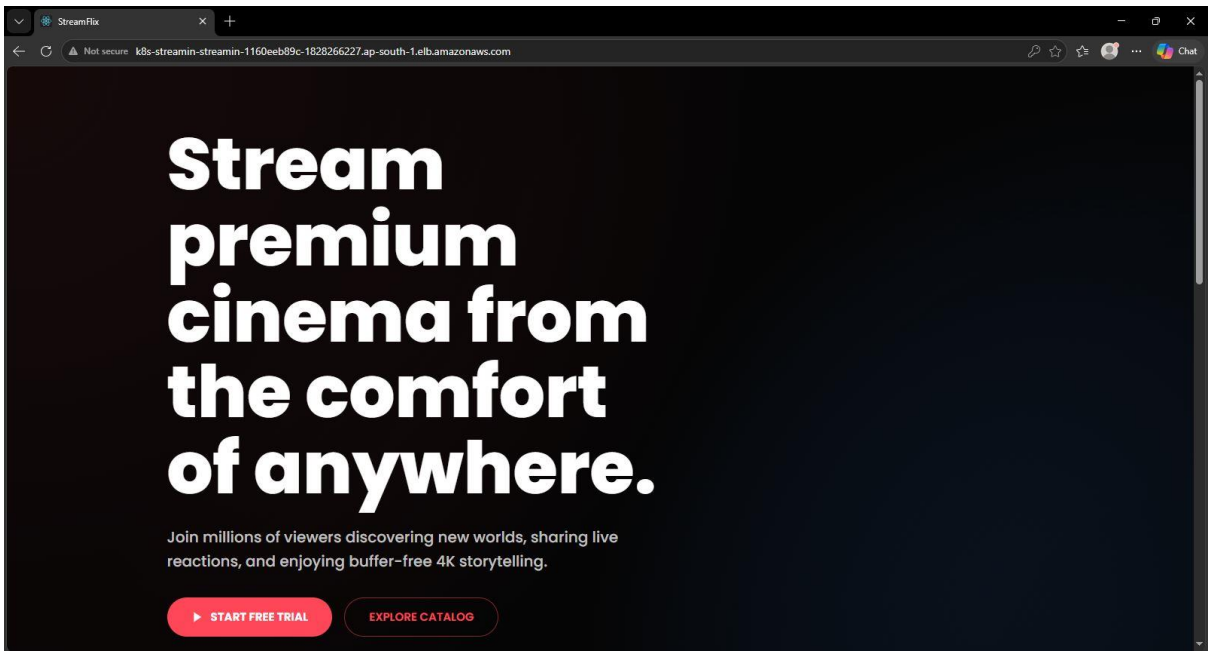
6. Ingress

```
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp> kubectl get ingress -n streaming
NAME           CLASS   HOSTS   ADDRESS                                                                  PORTS   AGE
streamingapp   alb     *       k8s-streamin-streamin-1160eeb89c-1828266227.ap-south-1.elb.amazonaws.com  80      80m
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp>
```

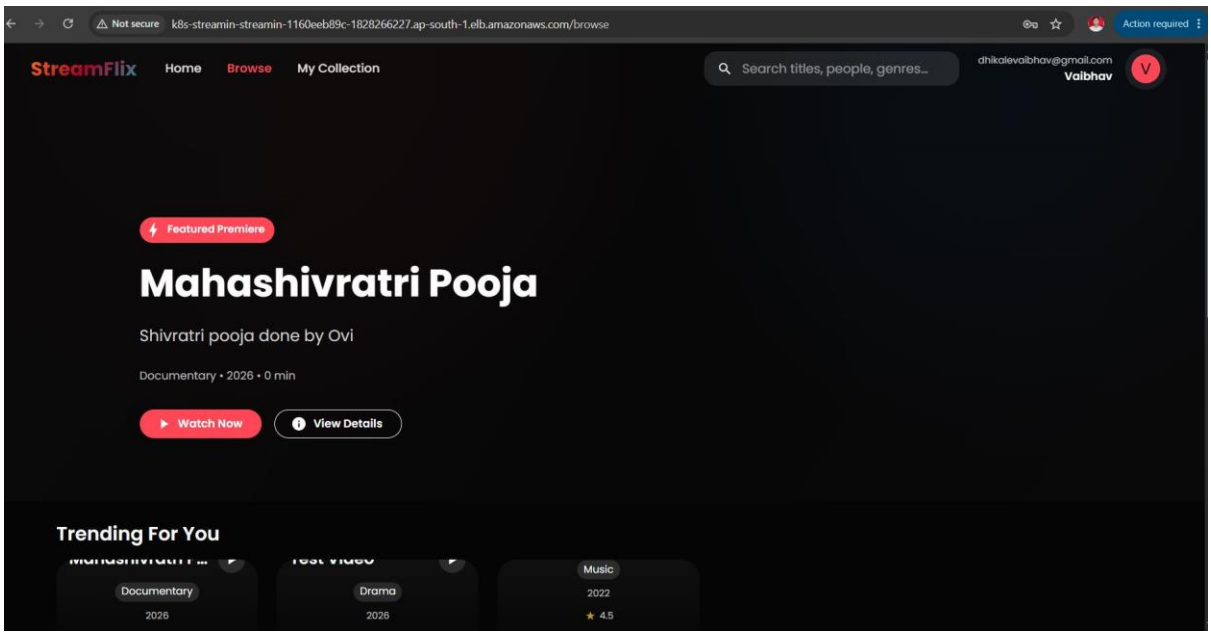
7. Helm

```
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp> helm list -n streaming
NAME            NAMESPACE       REVISION        UPDATED                                 STATUS          CHART                   APP VERSION
streamingapp    streaming       3               2026-10-07 15:12:02.7330427 +0530 IST   deployed        streamingapp-0.1.0      1.16.0
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp>
```

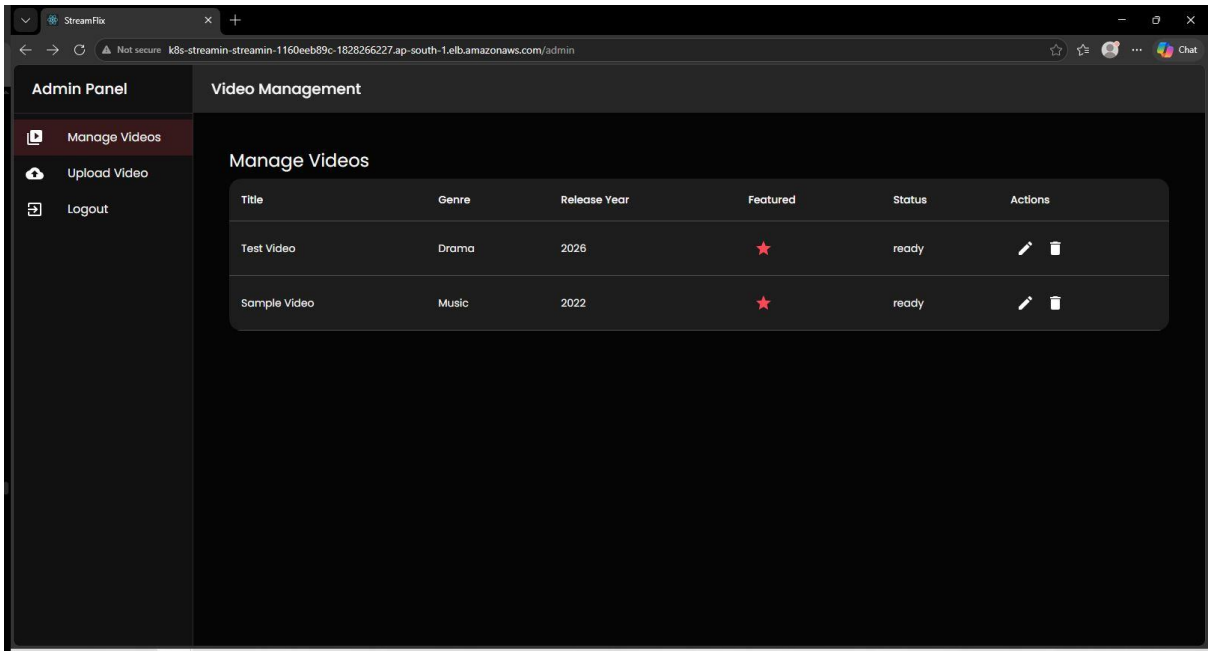
8. Running Application



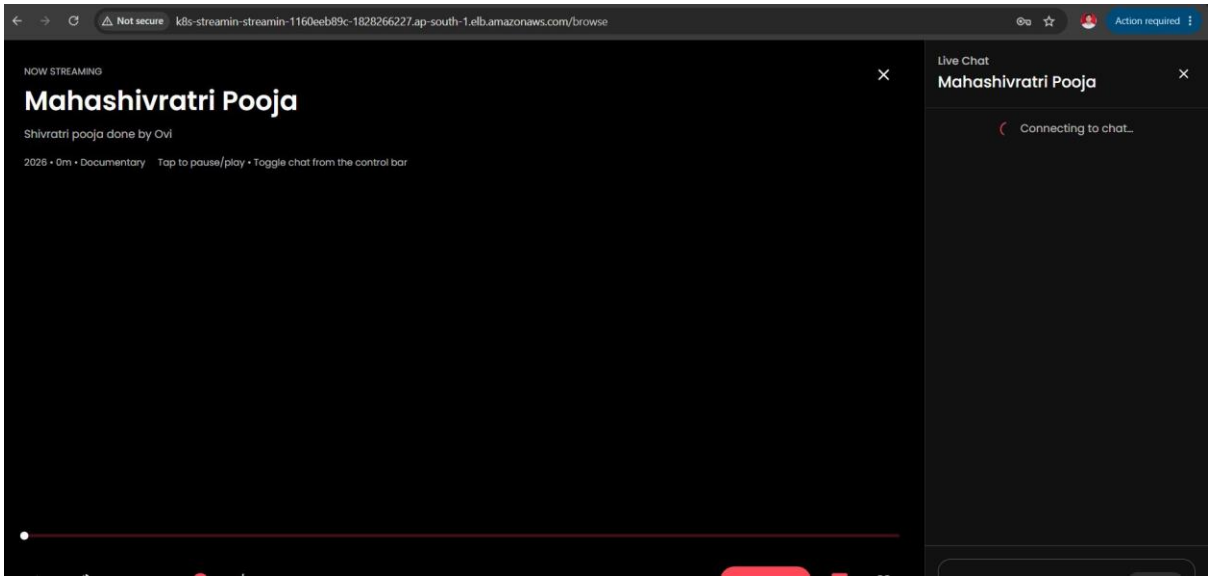
9. Login



10. Upload



11. Chat



12. Scaling

```
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp> kubectl get pods -n streaming
NAME                                    READY   STATUS    RESTARTS   AGE
mongo-0                                 1/1     Running   0          2d
streamingapp-admin-6c85cff768-r46qm     1/1     Running   0          75m
streamingapp-auth-7bcc5dcd5f-wgsxf      1/1     Running   0          14m
streamingapp-chat-6d6b6bb854-kvpx6      1/1     Running   0          2d
streamingapp-frontend-778f844f69-s96tp  1/1     Running   0          3h57m
streamingapp-streaming-979b65c57-8c786  1/1     Running   0          30m
streamingapp-streaming-979b65c57-8jbsl  1/1     Running   0          30m
streamingapp-streaming-979b65c57-h6s22  1/1     Running   0          16h
streamingapp-streaming-979b65c57-ktrhc  1/1     Running   0          30m
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp>
```

13. Rolling update

```
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp> kubectl rollout status deployment/streamingapp-auth -n streaming
deployment "streamingapp-auth" successfully rolled out
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp>
```

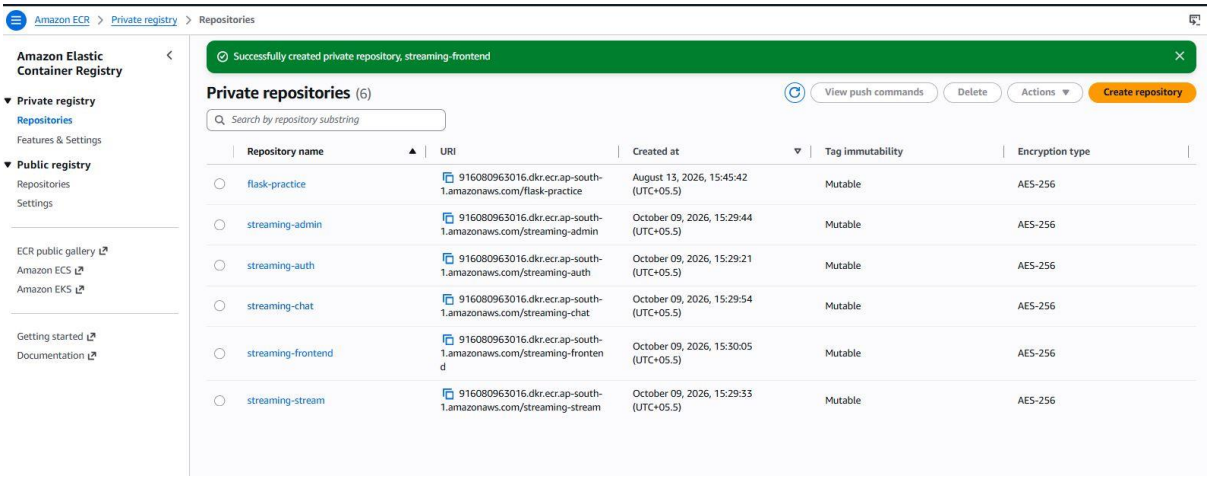
14. Self-healing

```
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp> kubectl get pods -n streaming
NAME                                    READY   STATUS    RESTARTS   AGE
mongo-0                                 1/1     Running   0          2d
streamingapp-admin-6c85cff768-r46qm     1/1     Running   0          59m
streamingapp-auth-7bcc5dcd5f-jp4b6      1/1     Running   0          22h
streamingapp-chat-6d6b6bb854-kvpx6      1/1     Running   0          2d
streamingapp-frontend-778f844f69-s96tp  1/1     Running   0          3h41m
streamingapp-streaming-979b65c57-8c786  1/1     Running   0          15m
streamingapp-streaming-979b65c57-8jbsl  1/1     Running   0          15m
streamingapp-streaming-979b65c57-h6s22  1/1     Running   0          15h
streamingapp-streaming-979b65c57-ktrhc  1/1     Running   0          15m
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp> kubectl delete pod streamingapp-auth-7bcc5dcd5f-jp4b6 -n streaming
pod "streamingapp-auth-7bcc5dcd5f-jp4b6" deleted from streaming namespace
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp> kubectl get pods -n streaming
NAME                                    READY   STATUS    RESTARTS   AGE
mongo-0                                 1/1     Running   0          2d
streamingapp-admin-6c85cff768-r46qm     1/1     Running   0          61m
streamingapp-auth-7bcc5dcd5f-wgsxf      1/1     Running   0          11s
streamingapp-chat-6d6b6bb854-kvpx6      1/1     Running   0          2d
streamingapp-frontend-778f844f69-s96tp  1/1     Running   0          3h43m
streamingapp-streaming-979b65c57-8c786  1/1     Running   0          16m
streamingapp-streaming-979b65c57-8jbsl  1/1     Running   0          16m
streamingapp-streaming-979b65c57-h6s22  1/1     Running   0          15h
streamingapp-streaming-979b65c57-ktrhc  1/1     Running   0          16m
PS D:\DevOps\Assignments\Graded Project on Orchestration and Scaling\StreamingApp>
```

15. Jenkins



16. ECR



17.CloudWatch

